Milestone: (E) ER and UML diagram

Group 4

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Percentage of Effort

Contributed by Student1: 50%

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Smart Food Delivery & Marketplace Management System

Dev Patel and Hitaxi Lethwala

As cities grow busier, food delivery has become an important part of everyday life. Customers, restaurants, and delivery drivers are all connected through online platforms. To keep everything running smoothly, there needs to be a smart system that can manage orders, deliveries, payments, discounts, and customer feedback. A well-structured database can make these daily tasks easier, more organized, and efficient. In this project, we will design a food delivery platform from the company's perspective. The system will show how the company handles customers, restaurants, menus, orders, drivers, vehicles, payments, and promotions to work more effectively and increase profits.

Entities and Requirements

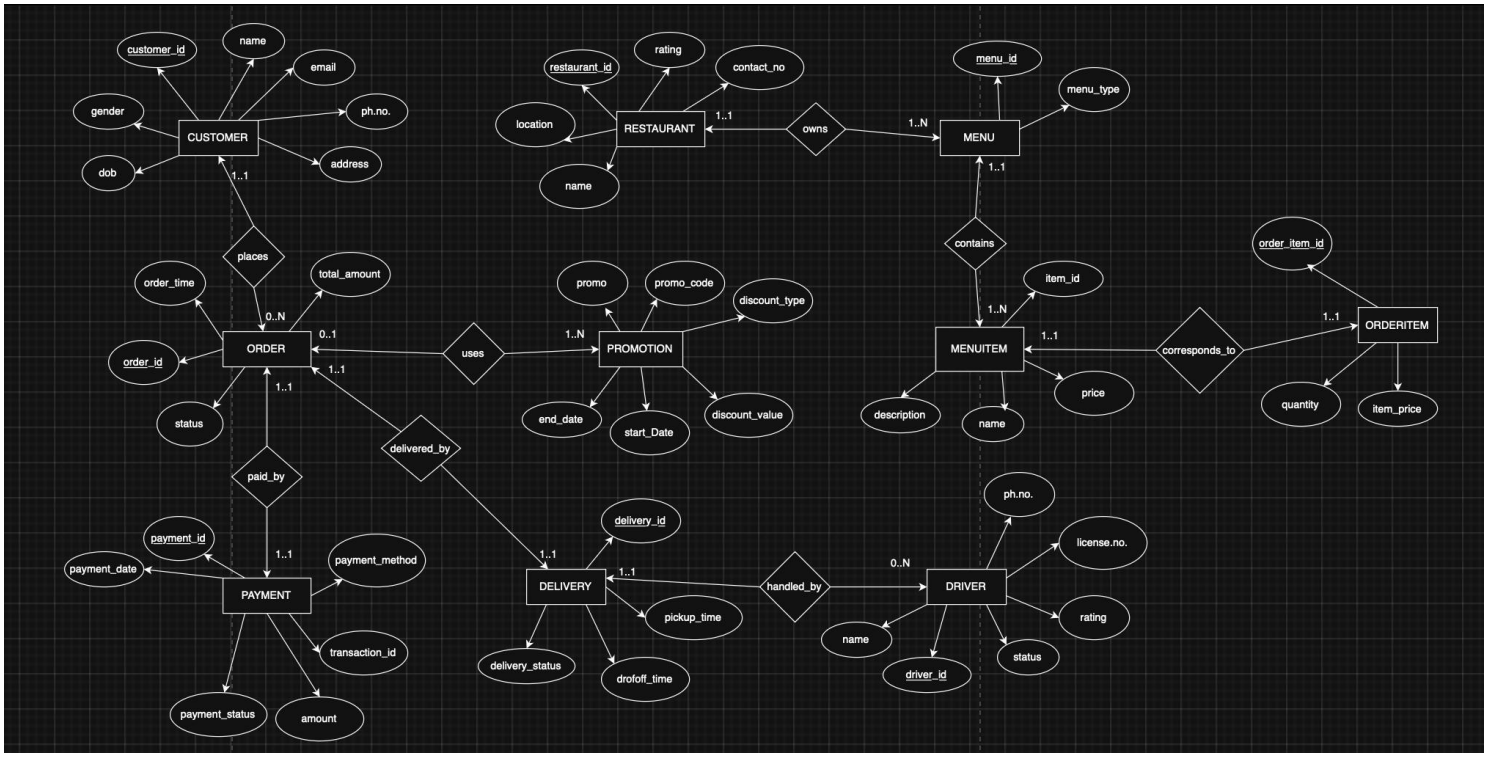
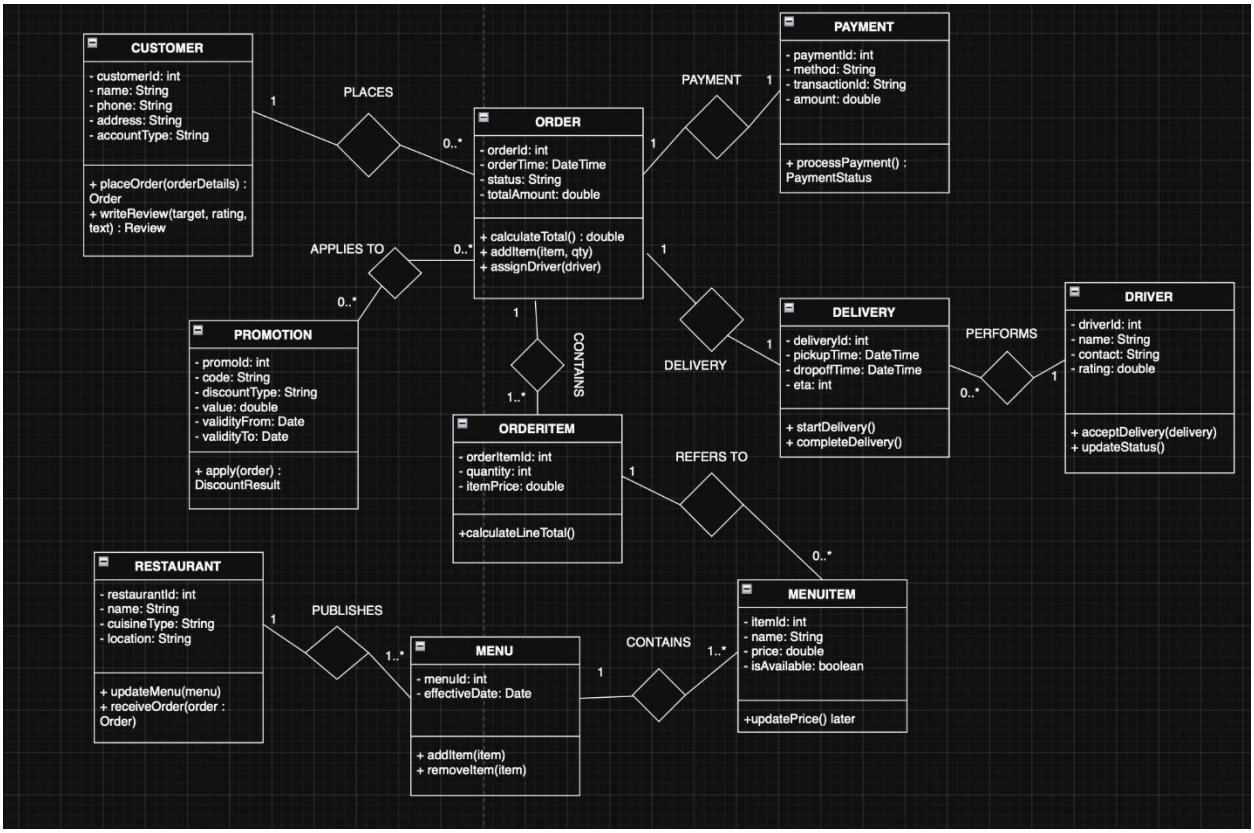
The company needs to keep track of the following information:

- **Customer:** customer_id, name, gender, date of birth, phone number, address, account_type.
- **Restaurant:** restaurant_id, name, cuisine_type, location, rating.
- **Menu:** menu_id, restaurant_id, effective_date.
- **Menu Item:** item_id, menu_id, name, price, availability.
- **Order:** order_id, customer_id, restaurant_id, order_time, total_amount, status.
- **Order Item:** order_item_id, order_id, item_id, quantity, item_price.
- **Payment:** payment_id, order_id, method, transaction_id, amount, status.
- **Driver:** driver_id, name, rating, contact, status.
- **Vehicle:** vehicle_id, driver_id, type, license_plate.
- **Delivery:** delivery_id, order_id, driver_id, pickup_time, dropoff_time, eta, status.
- **Review:** review_id, reviewer_id, target_type, rating, comment.
- **Promotion:** promo_id, code, discount_type, value, validity.

Other Requirements

1. A customer can place zero to many orders; each order belongs to one customer.
2. A menu must have at least one item; each item belongs to one menu.
3. An order can include multiple items; each item belongs to one order.
4. Each order generates one payment; a payment belongs to one order.
5. Each delivery belongs to one order; one driver can handle many deliveries.
6. A driver can have zero or one vehicle; each vehicle belongs to one driver.
7. A customer can write zero to many reviews; each review belongs to one customer.
8. A promotion can be used in many orders; an order can use zero or one promotion.

(E) ER Diagram:



UML Diagram: